INFO-I590: Fundamentals and Applications of LLMs

Overview and Introduction

Jisun An

Reuters

OpenAI says ChatGPT's weekly users have grown to 200 million

AI startup OpenAI said on Thursday its chatbot ChatGPT now has more than 200 million weekly active users, double the number it had in the...

Aug 29, 2024

TechCrunch

Sam Altman says ChatGPT has hit 800M weekly active users

Sam Altman says ChatGPT has hit 800M weekly active users ... OpenAI CEO Sam Altman said Monday that ChatGPT has reached 800 million weekly active...

Oct 6, 2025

Fierce Healthcare

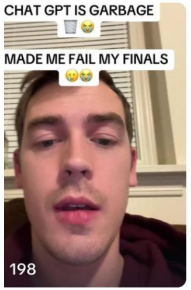
40M people use ChatGPT to get answers to healthcare questions, OpenAI says

Of the more than 800 million regular users of ChatGPT, 1 in 4 submits a prompt about healthcare every week, according to OpenAI.

1 week ago

source: https://www.reuters.com/technology/artificial-intelligence/openai-says-chatgpts-weekly-users-have-grown-200-million-2024-08-29/, https://techcrunch.com/2025/10/06/sam-altman-says-chatgpt-has-hit-800m-weekly-active-users/
https://www.fiercehealthcare.com/ai-and-machine-learning/40m-people-use-chatgpt-answer-healthcare-questions-openai-says

Chat Gpt Down for Finals Week



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studyhacks... ▷ 3589

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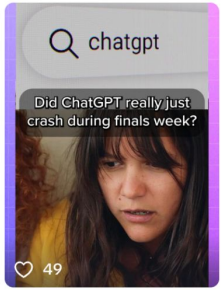

ChatGPT and Sora Go...
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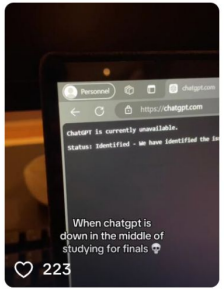
its looking like its you...
maryyy082 ▷ 137.9K



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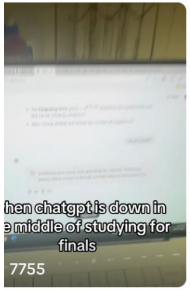

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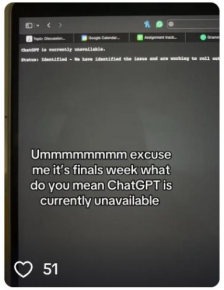

What's going on...
kat.ingle ▷ 4577

Whooooo gon help me...
_jocelynmo... ▷ 1033

Lit #finals #finalsweek...
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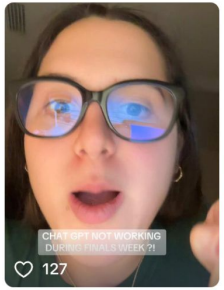


Going to have a mental...
boujeeswee... ▷ 2442

img: https://status.openai.com/history, https://www.tiktok.com/discover/is-chat-gpt-down-for-finals-week

The Top 50 Gen AI Web Products, by Unique Monthly Visits

1. ChatGPT	11. Kimi	21. CIVITAI	31. Photoroom	41. Monica
2. deepseek	12. Hailuo AI	22. IIElevenLabs	32. Moescape AI	42. CURSOR
3. character.ai	13. Hugging Face	23. Sora	33. Midjourney	43. ideogram
4. perplexity	14. Poe	24. Crushon AI	34. candy.ai	44. CHUB
5. JanitorAI	15. Adot	25. BLACKBOX AI	35. zeemo	45. Clipchamp
6. Claude	16. Eden AI	26. DeepAI	36. VEED	46. Meta AI
7. QuillBot	17. PolyBuzz	27. Gamma	37. invideo AI	47. StudyX
8. SUNO	18. SEAART.AI	28. Leonardo.Ai	38. Pixelcut	48. bolt
9. SPICYCHAT.AI	19. liner	29. cutout.pro	39. talkie	49. PicWish
10. Doubao	20. KLING AI	30. BRAINLY	40. PixAI	50. Joyland

Source: Similarweb, January 2025

Charts are for informational purposes only. Past performance is not indicative of future results. None of the above should be taken as investment advice; see a16z.com/disclosures.

a16z Consumer

img: https://a16z.com/100-gen-ai-apps-4/

What can we you do with LLMs?

- Help with writing and editing (e.g., grammar checks, emails, etc)
- Coding
- Knowledge searching, learning new concept
- Translation
- Travel plan
- Medical advice
- …

What are people using LLMs for?

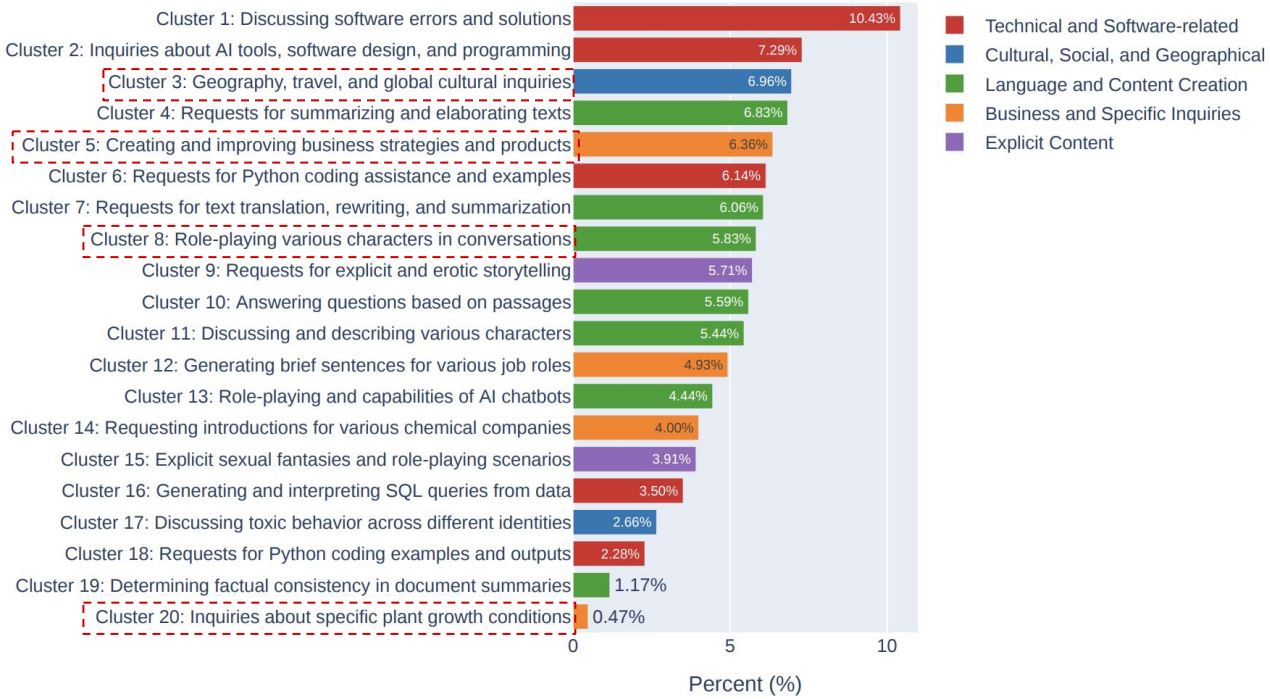


Figure 3: Topic distribution of 100K sampled conversations. Manual inspection of cluster centroids

https://arxiv.org/pdf/2309.11998

LLMs may fail significantly…

AI hallucinations caused artificial intelligence to falsely describe these people as criminals

By Anna Kelsey-Sugg and Damien Carrick for Law Report

ABC RN | Artificial Intelligence

Sun 3 Nov

DIVE BRIEF

Legal GenAI tools mislead 17% of time: Stanford study

Specialized legal applications are better than a general purpose tool like ChatGPT but whether they save time, given the oversight they need, is a question, an academic study says.

Published May 24, 2024

TechCrunch

Hacker tricks ChatGPT into giving out detailed instructions for making homemade bombs

The New York Times

Can A.I. Be Blamed for a Teen's Suicide?

The mother of a 14-year-old Florida boy says he beca
with a chatbot on Character.AI before his de

NBC NEWS

Lawsuit claims Character.AI is responsible for teen's suicide

Megan Garcia says the company's chatbots encouraged her 14-year-old son, Sewell Setzer, to take his own life, according to the lawsuit.

"They were all court cases I wrote about," Bernklau says.

NBC NEWS WATCH

Adam confided to ChatGPT

Adam confided to ChatGPT

ChatGPT pulled Adam deeper into a dark and hopeless place

Adam confided to ChatGPT

Courtesy Raine Family, NBC News

ARTIFICIAL INTELLIGENCE

The family of teenager who died by suicide alleges OpenAI's ChatGPT is to blame

The parents of Adam Raine, who died by suicide in April, claim in a new lawsuit against OpenAI that the teenager used ChatGPT as his "suicide coach."

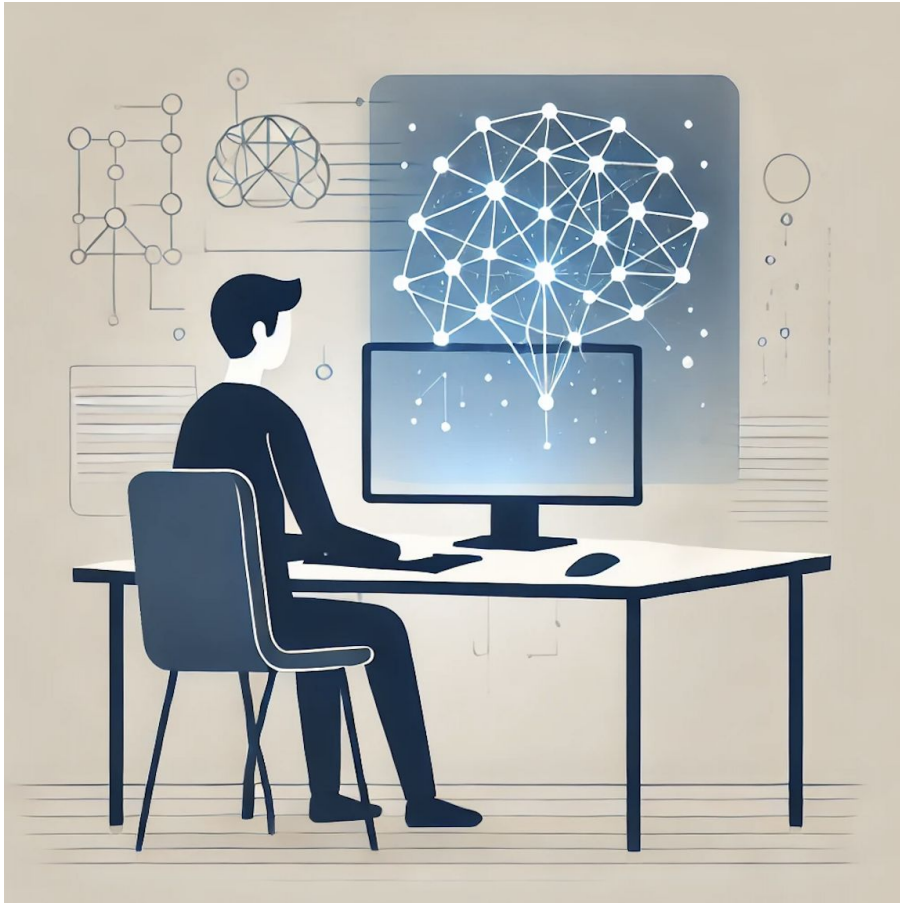
CNBC

AI tools such as ChatGPT are generating a mammoth increase in malicious phishing emails

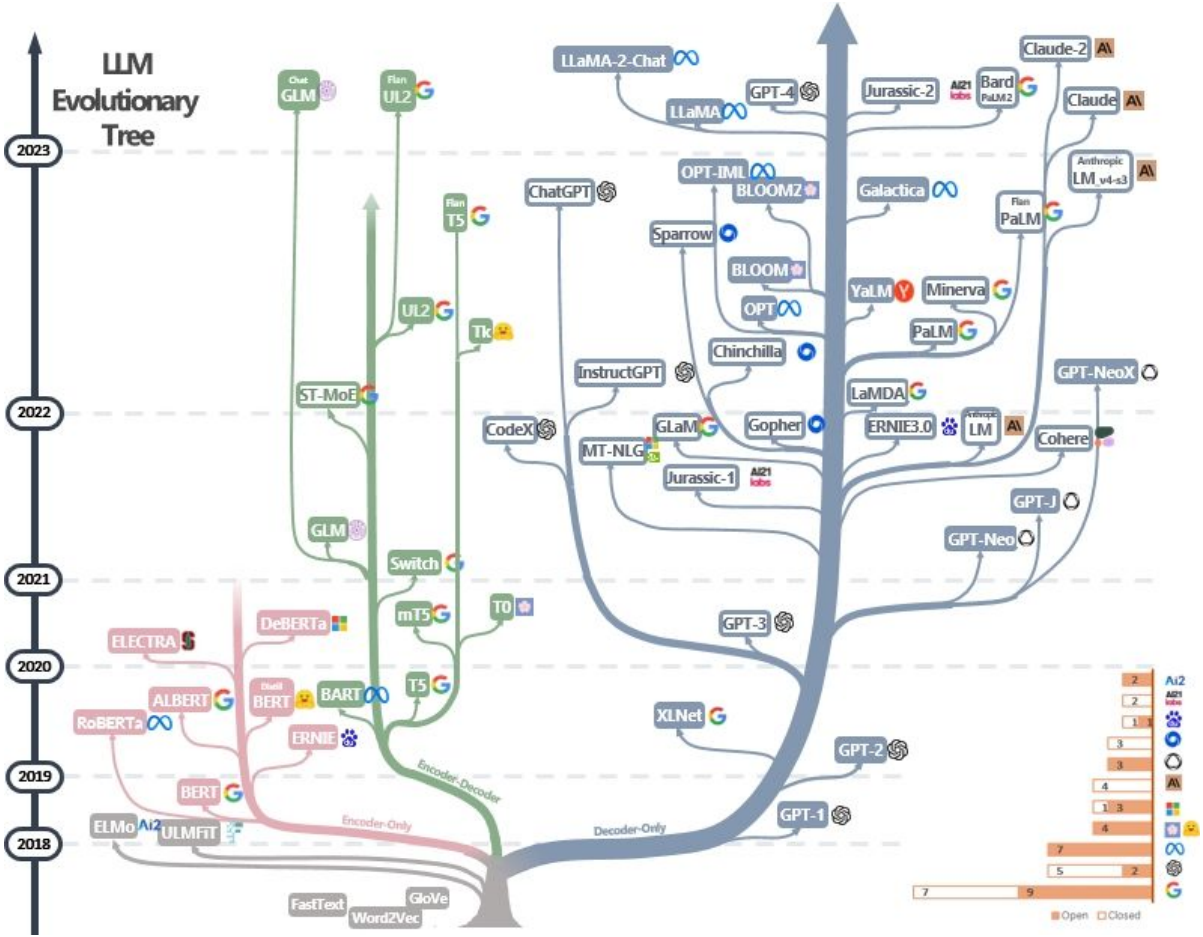
PUBLISHED TUE, NOV 28 2023•10:39 AM EST

Bob Violino

SHARE



Img: created by GPT-4o (feat. DALL·E 3)



Img: https://github.com/Mooler0410/LLMsPracticalGuide

Welcome to INFO-I590: Fundamentals and Applications of Large Language Models (LLMs)

What is a Large Language Model?

What is NLP (Natural Language Processing)?

- NLP is a form of artificial intelligence (AI) that aims to enable **computers** to **understand, interpret, and interact** with **human languages**, whether it be written, spoken, or even scribbled.

Applications of NLP

- Text Analysis and Understanding (e.g. sentiment analysis, topic modeling)
- Language Generation (e.g. chatbots, content creation)
- Code Generation
- Machine Translation
- Speech Recognition and Processing (e.g. speech-to-text)
- Information Retrieval (e.g. search engines)
- Named Entity Recognition (NER)
- Question Answering
- Text Classification

Techniques Used in NLP

- Rule-based Approaches
  - Using predefined linguistic rules to process language
- Statistical Methods
  - Leveraging probabilistic models to analyze language patterns (e.g.,n-grams)
- Machine Learning and Deep Learning
  - Training models like Support Vector Machines (SVM) or neural networks
  - Popular deep learning models include:
    - RNNs (Recurrent Neural Networks) and LSTMs for sequence data
    - **Transformers** (e.g., BERT, GPT) for large-scale language understanding and generation

Large Language Model (LLM)

- LLM is an **advanced type of NLP models** designed to understand, generate, and manipulate human language **by leveraging deep learning techniques**. LLMs are pre-trained on massive datasets from diverse text sources and are built on *transformer architectures.*
- Applications of LLMs?
  - LLMs can handle, and often excel in, most applications of NLP without requiring separate, specialized algorithms for each application.
  - Instead of creating individual models for specific NLP tasks, a single LLM can perform a wide range of tasks effectively.

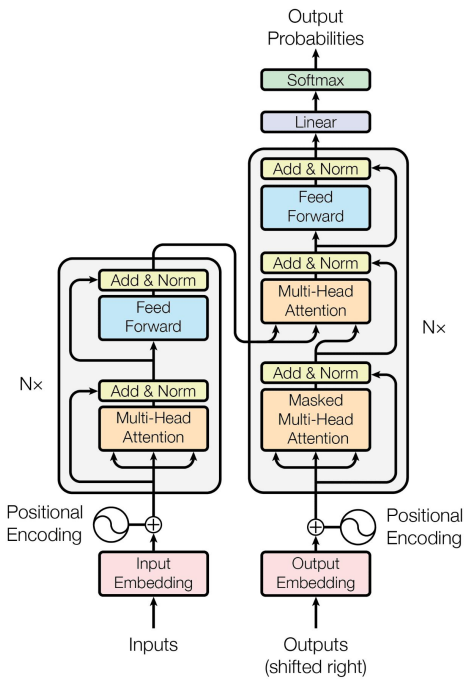
In this class, we will:

- Learn the principles, architectures, and training methods of large language models;
- Gain hands-on experience fine-tuning and applying LLMs across various domains;
- Critically assess the ethical, social, and practical implications of LLMs.

Roadmap

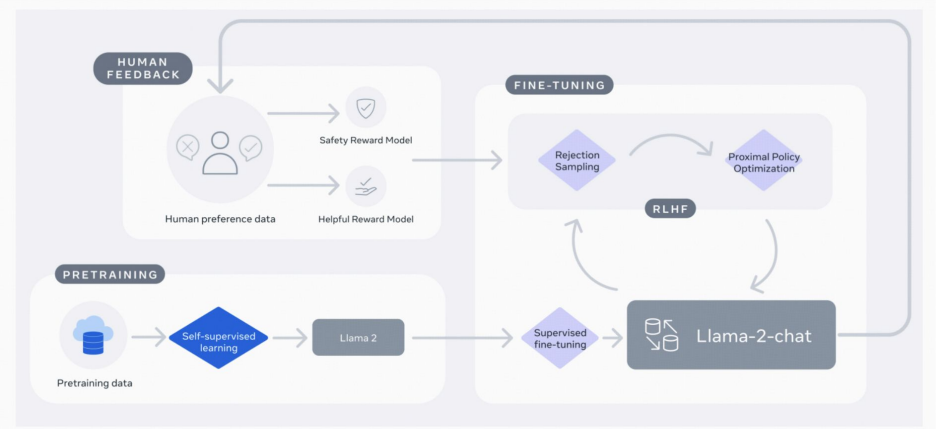
Topic 1: Language Modeling Fundamentals

- Representing words
- Language modeling
- Sequence encoding
- Transformers



Topic 2: Training and Inference Methods

- Pre-training
- Fine-tuning
- Instruction Tuning
- Prompting
- Reinforcement Learning



Img: LLama-2

Topic 3: LLM Applications

- Complex reasoning tasks
- Retrieval Augmented Generation
- Language agents

Topic 4: LLM in various domains

- Education
- Finance
- Business
- Law
- Health care
- Social science

Img: created by GPt-4o (feat DALL·E 2)

Class Format/Structure

Teaching Team

- Instructor: Jisun An
  - Office hours by appointment only
- TAs
  - Fan Huang
    - Office hours: Friday 3-4PM
    - Zoom: https://iu.zoom.us/my/fanhuang
  - Juyoung An
    - Office hours: Wednesday 2-3PM
    - Zoom: https://iu.zoom.us/my/iu.jy

Prerequisites

To succeed in this course, you are expected to:

- Be proficient in Python, which is required for project work;
- Have a basic understanding of linear algebra, probability, and statistics;
- Be familiar with basic concepts in NLP, such as train/dev/test splits, model fitting, features, and supervised learning. Please note that these topics will not be covered in this course.

Overview of Class Activities

- Group Project: 40%
  - Proposal presentation: 10%
  - Final Presentation: 10%
  - Final Submission: 20%
- Theoretical Assignments: 15%
- Practical Assignment: 15%
- Exam: 20%
- Participation: 10%

Attendance (10%)

- We will take attendance. You may **miss up to four classes without penalty**.

Canvas -> Assignments

-> Attendance -> Thur Jan 15

password:

llm

Theoretical Assignments (15%)

- There will be two individual assignments (T1 and T2, 7.5% each).
- Each assignment will require students to answer multiple-choice questions about the course content.

Practical Assignment (15%)

- One individual programming assignment using LLMs
- Details will be announced soon!

Exam (20%)

- Multiple-choice questions about the course content
- In-person, closed-book exam

Group Projects (40%)

- Scope
  - Applying LLMs to real-world problems for social good
  - Proposing new methods tailored to LLM challenges
  - Case studies of human factors in existing LLM systems
- 1-5 students per group
- Group Project
  - Proposal presentation (10%): 8 minutes
  - Final presentation (10%): 10 minutes
  - Final report (20%): 8 pages long (in ACL format)

Project examples (1)

- Mockmate - An Interview Bot
- TA Lite - Virtual Academic Assistant
- AI Driven Medical Chatbot
- Personalized Travel Planner Agent
- Multi agent based Deep Researcher
- Preventing Catastrophic Forgetting in LLMs
- Summarization on Social Media Post related to breast cancer treatment
- Customer Segmentation using Transformer-Based Models and Word2Vec Embeddings on Google Reviews

Project examples (2)

- LLM-Driven Multi-Turn Depression Screening Chatbot
- Card IQ: RAG-powered Multi-Agent System for Credit Card Selection and Optimization
- FinBuddy: An LLM-Powered Personal Finance Assistant
- FinSense: An LLM-Powered Financial News Impact Analyzer
- When Language Shape Life Stories: A Study of Simplified-Traditional Chinese and Bias in Large Language Models
- Can Vision-Language Models Do Reddit Moderation?

Schedule

Week	Topic	Remarks/Assessment
1	Jan 13 Tue - no class	
	Jan 15 Thu - Overview and introduction	
2	Jan 20 Tue - Word representations	
	Jan 22 Thu - Neural networks	
3	Jan 27 Tue - Language modeling	
	Jan 29 Thu - Lab: Text classification	
4	Feb 3 Tue - Attention and transformers	Assignment T1 released
	Feb 5 Thu - Pre-training and pre-trained models	
5	Feb 10 Tue - Fine-tuning and instruction-tuning	Assignment P1 released
	Feb 12 Thu - Prompting and reasoning	Feb 15 S-Submit T1 & Team formation
6	Feb 17 Tue - Lab - prompting	
	Feb 19 Thu - Lab - fine-tuning LLMs	
7	Feb 24 Tue - Reinforcement Learning from Human Feedback	
	Feb 26 Thu - Direct Preference Optimization, Group Relative Policy Optimization	
8	Mar 3 Tue - Retrieval augmented generation	Assignment T2 released
	Mar 5 Thu - Lab: RAG	Mar 9 M- Submit presentation slides

Schedule

9	Mar 10 Tue - Project proposal presentation	
	Mar 12 Thu - Project proposal presentation	Mar 15 S-Submit T2
10	Mar 17 Tue - Spring break, No class	
	Mar 19 Thu - Spring break, No class	
11	Mar 24 Tue - LLM agents (1)	
	Mar 26 Thu - LLM agents (2)	Mar 29 S- Submit P1
12	Mar 31 Tue - Lab: LLM agents	
	Apr 2 Thu - Exam	
13	Apr 7 Tue - AI safety	
	Apr 9 Thu - Vision LLMs	
14	Apr 14 Tue - Project hack week	
	Apr 16 Thu - Project hack week	Apr 20 M- Submit presentation slides
15	Apr 21 Tue - Project presentation	
	Apr 23 Thu - Project presentation	
16	Apr 28 Tue - Final report week (online, optional)	
	Apr 30 Thu - Final report week (online, optional)	May 3 S- Submit final report

Key Dates

Mark your calendar and plan ahead! Be sure to start the project early and plan ahead.

- Theoretical assignment 1 due: Feb 12 (Sun)
- Proposal presentation slide due: Mar 9 (Mon)
- Project proposal presentation: Mar 10 & 12 in class
- Theoretical assignment 2 due: Mar 8 (Sun)
- Spring break: Mar 17 & 19 (No class)
- Practical assignment 1 due: Mar 29 (Sun)
- **Exam: Apr 2 (Thur) in class**
- Final presentation slide due: Apr 20 (Mon)
- Final report submission: Mar 3 (Sun)

AI policy for Assignments / Projects

- AI tools are permitted in the work process, but the idea of your project should be your own.
- Transparency is crucial:
  - Declare which AI tools were used
  - Explain how they were employed
  - Specify tasks/sections and extent of AI use
- Include AI declaration in your report and be honest and specific about AI usage